

Ancient-data agent with source-audit feedback

Human-in-the-loop checks link archaeological context, fragile molecular signals, source-data QC, and decision feedback.

a. Archaeological source



Ancient sample record

Physical context and accession metadata define provenance and permissible downstream reuse.

b. Ancient DNA signal



ancient signal risks

● fragmented ● damaged ● contaminated

c. Agent-assisted audit

Source records
ENA/SRA metadata

FASTQ QC
read count · GC

Expert review
provenance

Audit outputs
reports

Decision gate

continue only when source files and QC match expectations

feedback: re-check metadata or file pairing
before downstream analysis

continue downstream

Output bundle: reports · intermediate files · workflow JSON · checksums · attribution

Message: ancient-material analysis should expose checkpoints where experts can stop, inspect, or request re-analysis.